



Observability Transformation

Cost · Performance · Operational Efficiency

Splunk / Datadog → Elastic



The "Burning Platform" — The Problem

- **"Success Tax": Costs grew FASTER than infrastructure**
 - Every spike in traffic = Permanent billing increase
- **Visibility Silos: Security & Engineering used different tools**
 - Finger-pointing during outages → slower resolution
- **Datadog High-Water Mark: 5-day spike billed for entire month**
- **Splunk Frozen Tier: 24+ hr rehydration — data effectively DARK**

COST vs INFRA GROWTH



Problem Deep Dive — Technical Pain Points



Billing Inefficiency

High-Water Mark billing punishes traffic spikes
You pay peak-month rates for an entire billing cycle



Data Accessibility

Splunk's Frozen Tier requires 24+ hour rehydration
Historical audit data is practically inaccessible



Fragmented Stack

Security & Engineering in separate tooling silos
No unified view = conflicting data during incidents



Operational Friction

Slow incident response due to siloed data
Manual troubleshooting without AIOps support

Strategic Decision — Why Elastic

STRATEGIC SHIFT

"Pay for Bytes" → "Pay for Resources"

From unpredictable data-volume billing to predictable resource-based pricing

- Search-first architecture — built beyond log parsing
- Platform consolidation: Security + Engineering on ONE stack
- Full observability: logs, metrics, traces, APM — unified
- Open-source core = no proprietary lock-in

Strategic Decision — Why Elastic

BEFORE vs AFTER

Dimension	Before	Elastic
Billing Model	Data volume	Resources
Data Access	24-hr rehydration	Instant (S3) with Searchable snapshots
Tooling	Multiple silos	Unified stack
Vendor Risk	Lock-in	Open / OTel
Historical Data	Frozen / dark	Always live

Game Changer — Searchable Snapshots



Legacy: Splunk Frozen Tier

Required expensive "Warm" hardware tier
Data recovery: 24+ hours — effectively DARK
Manual "thawing" process blocks audit workflows



Elastic: Searchable Snapshots

Data stored directly on S3 / Object Storage
Search results in sub-second to low-second
Years of history instantly accessible — no ops

24 hrs → 0

Rehydration Time Eliminated

COST DRIVER

HIGH

Splunk Warm
Hardware

LOW

Elastic S3
Storage

Standardization — ECS + OpenTelemetry



Elastic Common Schema (ECS)

Unified data format across firewalls, apps & clouds — one schema for everything



Open Telemetry (OTel)

Elastic contributed ECS to OTel — we are now vendor-neutral and future-proofed



No Vendor Lock-In

Standardized telemetry can move to any OTel-compatible platform — freedom guaranteed

DATA FLOW

Firewalls

Apps

Clouds

Endpoints



ECS



Elastic

Migration Strategy — Zero Risk Execution

1

Dual-Feed Setup

Legacy + Elastic receive identical data simultaneously

2

Query Conversion

SPL → ES|QL via Elastic AiOps

3

4-Week Validation

90%+ data continuity confirmed across all ingestion pipelines

4

Legacy Shutdown

Old licenses terminated only after full continuity sign-off

ZERO DOWNSIDE GUARANTEES

-  Zero alerts lost during migration
-  90%+ data continuity validated
-  4-week dual-feed safety window
-  Automated SPL → ES|QL conversion
-  No manual rewrite of query logic
-  Legacy licenses kept until full sign-off

Splunk / Datadog → Elastic

Transformation Results — Cost Reduction · Growth · Operational Efficiency

Results — Financial ROI

60%

Total Cost of Ownership Reduction

\$2.57

Return per \$1 Invested

14.5 mo

Average Payback Period

TCO COMPARISON — BEFORE vs AFTER



ROI BREAKDOWN

- Eliminated redundant SaaS licensing fees
- Removed expensive Warm hardware tier (Splunk)
- Regional data storage cuts 90% transfer costs
- Operational savings from unified tooling

Results — Engineering Productivity

BEFORE

- Manual troubleshooting workflows
- Alert fatigue from noisy dashboards
- Multiple tools = context switching

40%

Improvement in Debugging & Maintenance Efficiency

AFTER

- AIOps + Anomaly Detection automated
- Noise reduction → fewer false alerts
- Unified platform = instant context



AIOps Engine

Automated root cause detection reduces manual investigation time dramatically



Anomaly Detection

ML-driven alerting surfaces true incidents, cutting alert noise significantly



Unified Context

Security & Engineering share one platform — no tool-switching during incidents

Results — Performance Gains

30-80%

MTTR Reduction

Hours → Sec

Query Response Time

ZERO

Rehydration Bottlenecks

QUERY RESPONSE TIME — BEFORE vs AFTER

Historical Query

24+ Hours

< 5 Seconds

Live Log Search

15-30 min

< 1 Second

Incident Response

60+ min MTTR

10-18 min

After (Elastic)

Before (Legacy)



Future State — AI-Powered Observability

"We are no longer just Monitoring — we are Observing with full fidelity."



Elastic AI Assistant (Agent Builder)



Predictive Alerting (Anomaly)



99.9% Uptime



Shift: Monitor → Observe

Useful Links

<https://www.elastic.co/blog/migrating-from-splunk-to-the-elastic-elk-stack-data-migration>
<https://www.elastic.co/splunk-replacement>

Thank you — Questions Welcome